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Bandwidth consecutive multicolorings of graphs ☆

Kazuhide Nishikawa [a], Takao Nishizeki [a], Xiao Zhou [b,*]

[a] School of Science and Technology, Kwansei Gakuin University, 2-1 Gakuen, Sanda, 669-1337, Japan

[b] Graduate School of Information Sciences, Tohoku University, Sendai, 980-8579, Japan

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ABSTRACT

Let G be a simple graph in which each vertex v has a positive integer weight $b(v)$ and each edge (v, w) has a nonnegative integer weight $b(v, w)$. A bandwidth consecutive multicoloring of G assigns each vertex v a specified number $b(v)$ of consecutive positive integers so that, for each edge (v, w), all integers assigned to vertex v differ from all integers assigned to vertex w by more than $b(v, w)$. The maximum integer assigned to a vertex is called the span of the coloring. In the paper, we first investigate fundamental properties of such a coloring. We then obtain a pseudo polynomial-time exact algorithm and a fully polynomial-time approximation scheme for the problem of finding such a coloring of a given series-parallel graph with the minimum span. We finally extend the results to the case where a given graph G is a partial k-tree, that is, G has a bounded tree-width.

1. Introduction

An *ordinary coloring* of a graph G assigns each vertex a color so that, for each edge (v, w), the color assigned to v differs from the color assigned to w [7]. The problem of finding a coloring of a graph G with the minimum number $\chi(G)$ of colors often appears in the scheduling, task-allocation, etc. [7]. However, it is NP-hard, and difficult to find a good approximate solution. More precisely, for all $\varepsilon > 0$, approximating $\chi(G)$ within $n^{1-\varepsilon}$ is NP-hard [16], where n is the number of vertices in G.

The ordinary coloring has been extended in various ways [3–7,9,14,15]. A *multicoloring* assigns each vertex a specified number of colors so that, for each edge (v, w), the set of colors assigned to v is disjoint with the set of colors assigned to w [3–5,15]. A *bandwidth coloring* assigns each vertex a positive integer as a color so that the two integers assigned to the ends of each edge (v, w) differ by at least the specified weight $\omega(v, w)$ of (v, w) [9].

In this paper we deal with another generalized coloring, called a "bandwidth consecutive multicoloring". Let $G = (V, E)$ be a simple graph with vertex set V and edge set E. Each vertex $v \in V$ has a positive integer weight $b(v)$, while each edge $(v, w) \in E$ has a *non-negative* integer weight $b(v, w)$. A *bandwidth consecutive multicoloring* F of G is an assignment of positive integers to vertices such that

(a) each vertex $v \in V$ is assigned a set $F(v)$ of $b(v)$ consecutive positive integers; and
(b) for each edge $(v, w) \in E$, all integers assigned to v differ from all integers assigned to vertex w by *more than* $b(v, w)$.

We call such a bandwidth consecutive multicoloring F simply a *b-coloring* of G for a weight function b. The maximum integer assigned to a vertex is called the *span* of a b-coloring F, and is denoted by *span*(F). We define the *b-chromatic number* $\chi_b(G)$ of a graph G to be the minimum span over all b-colorings F of G. A b-coloring F is called *optimal* if *span*$(F) = \chi_b(G)$. A *b-coloring problem* is to compute $\chi_b(G)$ for a given graph G.

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* Corresponding author. Tel.: +81 22 795 7166.
E-mail addresses: nishikawa@kwansei.ac.jp (K. Nishikawa), nishi@kwansei.ac.jp (T. Nishizeki), zhou@ecei.tohoku.ac.jp (X. Zhou).

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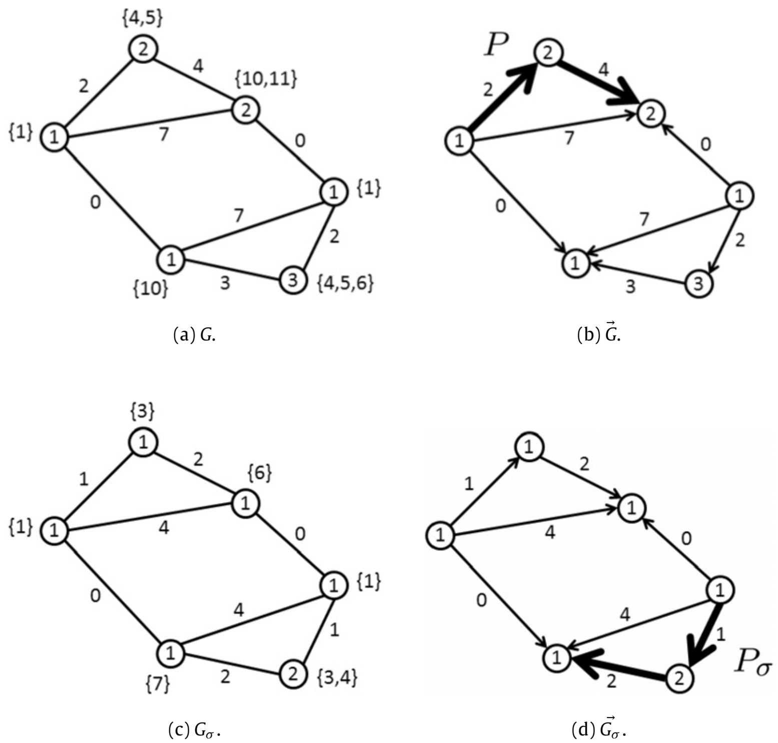

Fig. 1. (a) series-parallel weighted graph G and its optimal b-coloring F, (b) acyclic orientation $\vec{G}$ and the longest path P, (c) graph G_σ with weights scaled by $\sigma = 2$ and its optimal b_σ-coloring F_σ, and (d) acyclic orientation $\vec{G_\sigma}$ and the longest path P_σ.

Fig. 1(a) depicts a weighted graph G together with an optimal b-coloring F of G, where a weight $b(e)$ is attached to an edge e, a weight $b(v)$ is written in a circle representing a vertex v, and a set $F(v)$ is attached to a vertex v. Since $span(F) = 11$, $\chi_b(G) = 11$.

The ordinary vertex-coloring is merely a b-coloring for the case $b(v) = 1$ for every vertex v and $b(v, w) = 0$ for every edge (v, w). The "bandwidth coloring" or "channel assignment" [9] is a b-coloring for the case $b(v) = 1$ for every vertex v and $b(v, w) = \omega(v, w) - 1$ for every edge (v, w). It should be noted that our edge weight $b(v, w)$ is one less than the ordinary edge weight $\omega(v, w)$ of a bandwidth coloring. (This convention will make the arguments and algorithms simple and transparent.)

A b-coloring arises in the assignment of radio channels in cellular communication systems [9] and in the non-preemptive task scheduling [10]. The $b(v)$ consecutive integers assigned to a vertex v correspond to the contiguous bandwidth of a channel v or the consecutive time periods of a task v. The weight $b(v, w)$ assigned to edge (v, w) represents the requirement that the frequency band or time period of v must differ from that of w by more than $b(v, w)$. The span of a b-coloring corresponds to the minimum total bandwidth or the minimum makespan.

One can find a multicoloring of a graph G with the minimum number of colors in time polynomial in the output size if G is a series-parallel graph or a partial k-tree, that is, a graph of bounded tree-width [5,15]. The problem of finding a bandwidth coloring with the minimum number of colors is NP-hard even for partial 3-trees [9], and there is a fully polynomial-time approximation scheme (FPTAS) for the problem on partial k-trees [9]. Since our b-coloring problem is also NP-hard for partial 3-trees, it is desirable to obtain a good approximation algorithm. However, there are only heuristics for the b-coloring problem so far [8].

In this paper, we first investigate fundamental properties of a b-coloring. In particular, we characterize the b-chromatic number $\chi_b(G)$ of a graph G in terms of the longest path in acyclic orientations of G. We then obtain a pseudo polynomial-time exact algorithm for the b-coloring problem on series-parallel graphs, which often appear in the task scheduling and electrical circuits [10,12]. The algorithm takes time $O(B^3 n)$, where B is the maximum weight of G: $B = \max_{x \in V \cup E} b(x)$. Using the algorithm, we then give a fully polynomial-time approximation scheme (FPTAS) for the problem. We finally extend these results to the case where G is a partial k-tree. It should be noted that a series-parallel graph is a partial 2-tree. An early version of the paper was presented at a conference [11].

2. Preliminaries

In this section, we first give some definitions and then present three lemmas on a b-coloring.

Let $G = (V, E)$ be a simple graph without selfloops and multiple edges. We denote by n and m the number of vertices and edge in G, respectively. The *chromatic number* $\chi(G)$ of G is the minimum number of colors required by an ordinary coloring of G.

Let $\mathbb{N}$ be the set of all positive integers, that are regarded as colors. A *b-coloring* $F : V \to 2^{\mathbb{N}}$ of G must satisfy the following (a) and (b):

(a) for every vertex $v \in V$, the set $F(v)$ consists of $b(v)$ consecutive positive integers, and hence

$$\min F(v) = \max F(v) - b(v) + 1; \quad \text{and}$$

(b) for every edge $(v, w) \in E$, all integers in $F(v)$ differ from those in $F(w)$ by more than $b(v, w)$.

A b-coloring F can be represented by a function $f : V \to \mathbb{N}$ such that $f(v) = \max F(v)$ for every vertex $v \in V$. Clearly, for every vertex $v \in V$,

$$b(v) \le f(v). \tag{1}$$

For every edge $(v, w) \in E$,

$$f(v) \ne f(w) \tag{2}$$

since $b(v, w) \ge 0$. For every edge $(v, w) \in E$ with $f(v) < f(w)$,

$$b(v, w) < (f(w) - b(w) + 1) - f(v)$$

and hence

$$f(v) + b(v, w) + b(w) \le f(w). \tag{3}$$

Conversely, every function $f : V \to \mathbb{N}$ satisfying Eqs. (1)–(3) represents a b-coloring F such that

$$F(v) = \{f(v) - b(v) + 1, f(v) - b(v) + 2, \dots, f(v)\}.$$

Thus, such a function f is also called a *b-coloring* of G. Obviously, *span*$(F) = \max_{v \in V} f(v)$. We often denote *span*$(F)$ by *span*(f). A b-coloring f is called *optimal* if *span*$(f) = \chi_b(G)$. The *b-coloring problem* is to compute $\chi_b(G)$ for a given graph G with weight $b(x)$, $x \in V \cup E$.

The graph in Fig. 1(a) has the maximum weight $B = 7$. One can easily observe the following lemmas.

Lemma 2.1. *For every weighted graph $G = (V, E)$, $B \le \chi_b(G) \le B(2\chi(G) - 1)$.*

Proof. Obviously $B \le \chi_b(G)$. There is an ordinary coloring of G which uses a number $\chi(G)$ of colors c_i, $1 \le i \le \chi(G)$. Let $f : V \to \mathbb{N}$ be a function such that $f(v) = 2(i-1)B + b(v)$ if v is colored by c_i. Then f satisfies Eqs. (1)–(3), and hence f is a b-coloring of G. Therefore, $\chi_b(G) \le span(f) \le B(2\chi(G) - 1)$. □

Lemma 2.2. *Let $G = (V, E)$ be a bipartite graph in which every vertex has degree one or more, and let*

$$\bar{B} = \max\{b(v) + b(v, w) + b(w) \mid (v, w) \in E\}.$$

Then $\chi_b(G) = \bar{B}$.

Proof. Obviously $\bar{B} \le \chi_b(G)$. Thus, it suffices to prove that $\chi_b(G) \le \bar{B}$. Since G is a bipartite graph, G has an ordinary coloring with two colors c_1 and c_2. Define $f : V \to \mathbb{N}$ as follows:

$$f(v) = b(v) \quad \text{if } v \text{ is colored by } c_1;$$

and

$$f(w) = \bar{B} \quad \text{if } w \text{ is colored by } c_2.$$

Then f satisfies Eqs. (1)–(3), and hence f is a b-coloring of G. Hence $\chi_b(G) \le span(f) = \bar{B}$. □

Lemma 2.2 implies that the b-coloring problem can be solved in linear time for bipartite graphs and hence for trees.

We then characterize $\chi_b(G)$ in terms of the longest path in acyclic orientations of G. Orient all edges of G so that the resulting directed graph $\overrightarrow{G}$ is acyclic. The directed graph $\overrightarrow{G}$ is called an *acyclic orientation* of G. Fig. 1(b) depicts an acyclic orientation of the graph G in Fig. 1(a).

The length $\ell(P, D)$ of a directed path P in an acyclic graph D is the sum of the weights of all vertices and edges in P. We denote by $\ell_{\max}(D)$ the length of the longest directed path in D. For the acyclic graph $\overrightarrow{G}$ in Fig. 1(b) $\ell_{\max}(\overrightarrow{G}) = 11$, and the longest directed path P in $\overrightarrow{G}$ is drawn by thick lines.

Extending the Gallai–Roy theorem on the ordinary coloring (see for example [13]), we have the following lemma on the b-coloring.

Lemma 2.3. *For every graph $G = (V, E)$ with a weight function b*

$$\chi_b(G) = \min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G}),$$

where the minimum is taken over all acyclic orientations $\overrightarrow{G}$ of G.

Proof. We first prove that $\chi_b(G) \geq \min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G})$. Let f be an optimal b-coloring of G. Then $span(f) = \chi_b(G)$. Orient each edge $(v, w) \in E$ from v to w if and only if $f(v) < f(w)$. Then clearly the resulting directed graph D is acyclic. Let $P = v_1, e_1, v_2, e_2, \ldots, v_{p-1}, e_{p-1}, v_p$ be the longest directed path in D, where edge e_i, $1 \leq i \leq p-1$, goes from vertex v_i to v_{i+1}. Then

$$\ell_{\max}(D) = \sum_{i=1}^{p} b(v_i) + \sum_{i=1}^{p-1} b(v_i, v_{i+1}).$$

Since f is a b-coloring of G, by Eqs. (1) and (3) we have

$$b(v_1) \leq f(v_1) \tag{4}$$

and

$$b(v_i, v_{i+1}) + b(v_{i+1}) \leq f(v_{i+1}) - f(v_i) \tag{5}$$

for every i, $1 \leq i \leq p-1$. Taking the sum of Eqs. (4) and (5) for all i, $1 \leq i \leq p-1$, we have

$$\ell_{\max}(D) \leq f(v_p) \leq span(f) = \chi_b(G).$$

Since $\min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G}) \leq \ell_{\max}(D)$, we have $\chi_b(G) \geq \min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G})$.

We then prove that $\chi_b(G) \leq \min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G})$. Let D be an acyclic orientation of G such that

$$\ell_{\max}(D) = \min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G}).$$

Let $f : V \to \mathbb{N}$ be a mapping such that $f(v)$ is the length of the longest directed path in D ending at v for each vertex v of D. Then, for every directed edge (v, w) of D, $f(v) + b(v, w) + b(w) \leq f(w)$ and hence $f(v) \neq f(w)$. The definition of f implies that $b(v) \leq f(v)$ for every vertex $v \in V$. Thus f is a b-coloring of G and $span(f) = \ell_{\max}(D)$. Hence $\chi_b(G) \leq span(f) = \min_{\overrightarrow{G}} \ell_{\max}(\overrightarrow{G})$. $\square$

There are at most 2^m acyclic orientations of G, and one can compute $\ell_{\max}(\overrightarrow{G})$ in time $O(m+n)$ for each acyclic orientation $\overrightarrow{G}$ of G, where m and n are the numbers of edges and vertices in G, respectively. Thus, Lemma 2.3 implies that $\chi_b(G)$ can be computed in time $O((m+n)2^m)$, regardless of how large the weights are.

3. Exact algorithm for series-parallel graphs

Many problems can be solved for series-parallel graphs in polynomial time or even in linear time [12]. In this section we show that the b-coloring problem can be solved for series-parallel graphs in pseudo polynomial-time $O(B^3 n)$. It should be noted that $B^3 n$ is polynomial in n and B.

A *series-parallel graph* is recursively defined as follows [12]:

1. A graph G of a single edge is a series-parallel graph, and has the ends of the edge as terminals s and t of G. (See Fig. 2(a).)
2. Let G_1 be a series-parallel graph with terminals s_1 and t_1, and let G_2 be a series-parallel graph with terminals s_2 and t_2. (See Fig. 2(b).)
   (a) A graph G obtained from G_1 and G_2 by identifying t_1 with s_2 is a series-parallel graph, whose terminals are s_1 and t_2. Such a connection is called a *series connection*. (See Fig. 2(c).)
   (b) A graph obtained from G_1 and G_2 by identifying s_1 with s_2 and identifying t_1 with t_2 is a series-parallel graph, whose terminals are $s_1 = s_2$ and $t_1 = t_2$. Such a connection is called a *parallel connection*. (See Fig. 2(d).)

Every series-parallel graph G can be represented by a *"binary decomposition tree"*. Fig. 3 illustrates a decomposition tree T of the series-parallel graph G in Fig. 1(a). Labels **s** and **p** attached to internal nodes in T indicate series and parallel connections, respectively. Every leaf of T represents a subgraph of G induced by a single edge. A node u of T corresponds to a subgraph G_u of G induced by all edges represented by the leaves that are descendants of u in T. Thus $G = G_r$ for the root r of T. One can find a decomposition tree of a given series-parallel graph in linear time [12].

The definition immediately implies that every series-parallel graph G has an ordinary coloring with at most three colors, that is, $\chi(G) \leq 3$. Therefore, by Lemma 2.1, we have $\chi_b(G) \leq 5B$. For a series-parallel graph G with terminals s and t and

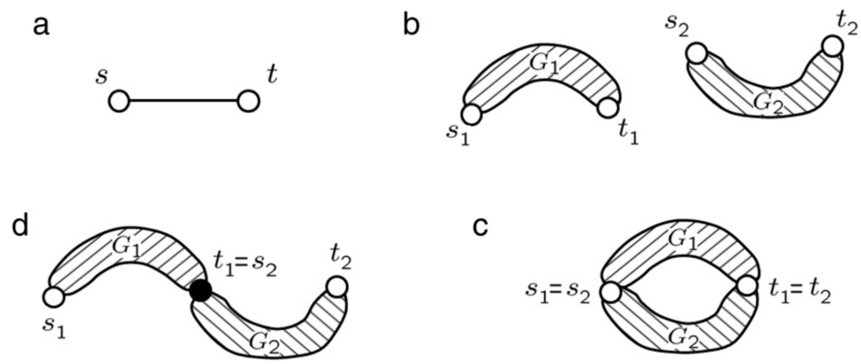

Fig. 2. Definition of series-parallel graphs.

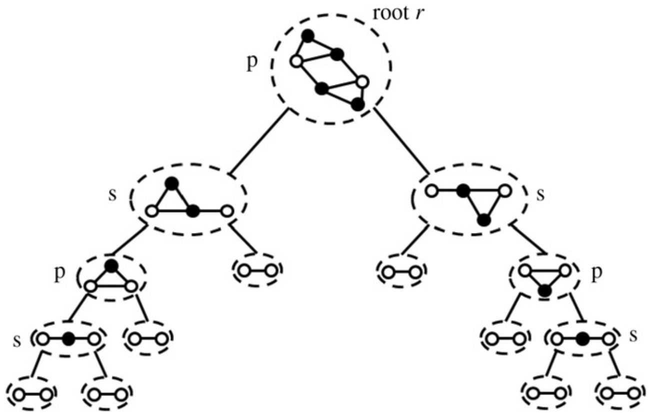

Fig. 3. Decomposition tree T of the series-parallel graph in Fig. 1(a).

integers i and j, $1 \leq i, j \leq 5B$, we define

$$\chi_{ij}(G) = \min_{f} span(f)$$

where the minimum is taken over all b-colorings f of G such that $f(s) = i$ and $f(t) = j$. Let $\chi_{ij}(G) = \infty$ if there is no such b-coloring.

One can recursively compute $\chi_{ij}(G)$, $1 \leq i, j \leq 5B$, as follows. Consider first the case where G consists of a single edge $e = (s, t)$ as illustrated in Fig. 2(a). Then $\chi_{ij}(G) = \max\{i, j\}$ if the following (a)–(c) hold:

(a) $i \neq j$, $b(s) \leq i$, and $b(t) \leq j$;
(b) $i < j$ implies $i + b(s, t) + b(t) \leq j$; and
(c) $j < i$ implies $j + b(s, t) + b(s) \leq i$.

Otherwise, $\chi_{ij}(G) = \infty$. Consider next the case where G is obtained from G_1 and G_2 by a series connection as illustrated in Fig. 2(c). Then

$$\chi_{ij}(G) = \min_{1 \leq k \leq 5B} \max\{\chi_{ik}(G_1), \chi_{kj}(G_2)\}. \tag{6}$$

Consider finally the case where G is obtained from G_1 and G_2 by a parallel connection as illustrated in Fig. 2(d). Then

$$\chi_{ij}(G) = \max\{\chi_{ij}(G_1), \chi_{ij}(G_2)\}. \tag{7}$$

One may assume that a series-parallel graph G has no multiple edges. Then one can easily prove by induction that $m \leq 2n - 3$. Since the binary decomposition tree T of G has m leaves, T has exactly $m - 1 (\leq 2n - 4)$ internal nodes. We compute $\chi_{ij}(G_u)$, $1 \leq i, j \leq 5B$, for all nodes u of T from leaves to the root r. It takes time $O(B^3 n)$. Since $G = G_r$, we compute $\chi_b(G)$ from $\chi_{ij}(G_r)$ in time $O(B^2)$ as follows:

$$\chi_b(G) = \min_{1 \leq i, j \leq 5B} \chi_{ij}(G_r).$$

Thus we have the following theorem.

Theorem 3.1. *The b-coloring problem can be solved in time $O(B^3 n)$ for a series-parallel graph G, where n is the number of vertices in G and B is the maximum weight of G.*

Clearly, $B^3 n$ is polynomial in n if B is bounded above by a polynomial in n.

4. FPTAS

In this section we give a fully polynomial-time approximation scheme (FPTAS) for the b-coloring problem on series-parallel graphs.

Let G be a graph with a weight function b, and let σ be a scaling factor which is a positive integer. Then we denote by G_σ a graph which is isomorphic with G but has a weight function b_σ such that

$$b_\sigma(x) = \lceil b(x)/\sigma \rceil \tag{8}$$

for every element $x \in V \cup E$. Fig. 1(c) depicts G_σ with $\sigma = 2$ for the graph G in Fig. 1(a). An optimal b_σ-coloring F_σ of G_σ is also depicted in Fig. 1(c). We now have the following lemma.

Lemma 4.1. *Let $G = (V, E)$ be a graph with a weight function b, let σ be a positive integer, and let f_σ be an optimal b_σ-coloring of G_σ. Then, a function f such that $f(v) = \sigma f_\sigma(v)$ for every vertex v is a b-coloring of G, and hence $\chi_b(G) \le \sigma \chi_{b_\sigma}(G_\sigma)$.*

Proof. Since f_σ is an optimal b_σ-coloring of G_σ, we have $span(f_\sigma) = \chi_{b_\sigma}(G_\sigma)$, $b_\sigma(v) \le f_\sigma(v)$ for every vertex $v \in V$, $f_\sigma(v) \ne f_\sigma(w)$ for every edge (v, w), and $f_\sigma(v) + b_\sigma(v, w) + b_\sigma(w) \le f_\sigma(w)$ for every edge (v, w) with $f_\sigma(v) < f_\sigma(w)$. Therefore, we have $b(v) \le \sigma b_\sigma(v) \le \sigma f_\sigma(v) = f(v)$ for every vertex v. Similarly, we have $f(v) \ne f(w)$ for every edge (v, w), and $f(v) + b(v, w) + b(w) \le f(w)$ for every edge (v, w) with $f(v) < f(w)$. Thus f is a b-coloring of G, and hence $\chi_b(G) \le span(f) = \sigma \cdot span(f_\sigma) = \sigma \chi_{b_\sigma}(G_\sigma)$. □

Consider the following approximation scheme.

Approximation scheme

1. Choose a scaling factor σ appropriately. (We will later choose $\sigma = \lfloor \varepsilon B/4n \rfloor$ for a desired approximation error rate ε.)
2. Find an optimal b_σ-coloring f_σ of G_σ (by a pseudo polynomial-time exact algorithm, say the algorithm in Section 3).
3. Output, as an approximate solution, a b-coloring f of G such that $f(v) = \sigma f_\sigma(v)$ for every vertex v.

We now have the following lemma on the longest paths in acyclic orientations $\vec{G}$ and $\vec{G_\sigma}$.

Lemma 4.2. *Let $G = (V, E)$ be a weighted graph of n vertices, let $\vec{G}$ be an acyclic orientation of G, and let P be the longest directed path in $\vec{G}$. Let σ be a positive integer, let $\vec{G_\sigma}$ be the acyclic graph obtained from G_σ by orienting each edge in the same direction as in $\vec{G}$, and let P_σ be the longest directed path in $\vec{G_\sigma}$. (See Fig. 1.) Then*

$$\ell(P_\sigma, \vec{G_\sigma}) < \ell(P, \vec{G_\sigma}) + 2n. \tag{9}$$

Proof. By Eq. (8) we have $b_\sigma(x) < b(x)/\sigma + 1$ for every element $x \in V \cup E$. Clearly there are at most $2n - 1$ elements (vertices and edges) in P_σ. Therefore, we have

$$\begin{aligned} \sigma \ell(P_\sigma, \vec{G_\sigma}) &= \sigma \sum_{x \in P_\sigma} b_\sigma(x) \\ &< \sigma \sum_{x \in P_\sigma} \left(\frac{b(x)}{\sigma} + 1 \right) \\ &\le \sum_{x \in P_\sigma} b(x) + \sigma(2n - 1) \\ &< \ell(P_\sigma, \vec{G}) + 2\sigma n \end{aligned} \tag{10}$$

where the summation is taken over all elements x in P_σ. Since P is the longest path in $\vec{G}$,

$$\ell(P_\sigma, \vec{G}) \le \ell(P, \vec{G}). \tag{11}$$

Since $b(x) \le \sigma b_\sigma(x)$ for every element $x \in V \cup E$, we have

$$\begin{aligned} \ell(P, \vec{G}) &= \sum_{x \in P} b(x) \\ &\le \sigma \sum_{x \in P} b_\sigma(x) \\ &= \sigma \ell(P, \vec{G_\sigma}). \end{aligned} \tag{12}$$

From Eqs. (10)–(12) we have $\sigma \ell(P_\sigma, \vec{G_\sigma}) < \sigma \ell(P, \vec{G_\sigma}) + 2\sigma n$. We have thus proved Eq. (9). □

Using Lemmas 2.3 and 4.2, we then have the following lemma on the error of the approximation scheme above.

Lemma 4.3. *For a positive integer σ and a graph $G = (V, E)$ with a weight function b*

$$\sigma \chi_{b_\sigma}(G_\sigma) < \chi_b(G) + 4\sigma n.$$

Proof. Lemma 2.3 implies that there is an acyclic orientation $\overrightarrow{G}$ of G and the longest path P in $\overrightarrow{G}$ such that

$$\chi_b(G) = \ell_{\max}(\overrightarrow{G}) = \ell(P, \overrightarrow{G}). \tag{13}$$

Let $\overrightarrow{G_\sigma}$ be the acyclic orientation obtained from G_σ by orienting each edge in the same direction as in $\overrightarrow{G}$. The path P contains at most $2n - 1$ elements (vertices and edges). Therefore, we have

$$\begin{aligned} \chi_b(G) + 2\sigma n &= \sum_{x \in P} b(x) + 2\sigma n \\ &> \sigma \sum_{x \in P} \left(\frac{b(x)}{\sigma} + 1 \right) \\ &> \sigma \sum_{x \in P} b_\sigma(x) \\ &= \sigma \ell(P, \overrightarrow{G_\sigma}). \end{aligned} \tag{14}$$

Let P_σ be the longest path in $\overrightarrow{G_\sigma}$, then by Lemma 4.2 we have

$$\ell(P, \overrightarrow{G_\sigma}) + 2n > \ell(P_\sigma, \overrightarrow{G_\sigma}). \tag{15}$$

By Lemma 2.3 we have

$$\ell(P_\sigma, \overrightarrow{G_\sigma}) \geq \chi_{b_\sigma}(G_\sigma). \tag{16}$$

By Eqs. (14)–(16) we have

$$\begin{aligned} \chi_b(G) + 4\sigma n &> \sigma \ell(P, \overrightarrow{G_\sigma}) + 2\sigma n \\ &> \sigma \ell(P_\sigma, \overrightarrow{G_\sigma}) \\ &\geq \sigma \chi_{b_\sigma}(G_\sigma). \quad \square \end{aligned}$$

Let ε (>0) be a desired approximation error rate. If $\varepsilon B/4n \leq 1$, then we compute $\chi_b(G)$ by a pseudo polynomial-time exact algorithm, say the algorithm in Section 3; the computation time is bounded by a polynomial in n and $1/\varepsilon$ since $B \leq 4n/\varepsilon$. One may thus assume that $\varepsilon B/4n > 1$. We then choose $\sigma = \lfloor \varepsilon B/4n \rfloor (\geq 1)$, and find an approximately optimal b-coloring $f\,(=\sigma f_\sigma)$ of G by the approximation scheme above. By Lemmas 2.1 and 4.3 one can bound the error as follows:

$$\begin{aligned} span(f) - \chi_b(G) &= \sigma \chi_{b_\sigma}(G_\sigma) - \chi_b(G) \\ &< 4\sigma n \\ &\leq \varepsilon B \\ &\leq \varepsilon \chi_b(G). \end{aligned} \tag{17}$$

We thus have the following theorem.

Theorem 4.1. *If there is an exact algorithm to solve the b-coloring problem for a class of graphs in time polynomial in n and B, then there is a fully polynomial-time approximation scheme for the class.*

Proof. Suppose that the algorithm finds an optimal b-coloring of a graph G in the class in time $p(n, B)$, where $p(n, B)$ is a polynomial in n and B. Find an optimal b_σ-coloring f_σ of G_σ in time $p(n, B_\sigma)$ by the algorithm, and output a b-coloring $f = \sigma f_\sigma$ of G by the approximation scheme above, where $\sigma = \lfloor \varepsilon B/4n \rfloor$ and $B_\sigma = \lceil B/\sigma \rceil$ is the maximum weight of G_σ. By Eq. (17) the error is less than $\varepsilon \chi_b(G)$. Since $B_\sigma = O(n/\varepsilon)$, the computation time $p(n, B_\sigma)$ of the scheme is bounded by a polynomial in n and $1/\varepsilon$. $\square$

From Theorems 3.1 and 4.1 we thus have the following corollary.

Corollary 4.1. *There is a fully polynomial-time approximation scheme for the b-coloring problem on series-parallel graphs, and the computation time is $O({B_\sigma}^3 n) = O(n^4/\varepsilon^3)$.*

5. Partial *k*-Trees

The class of partial k-trees, that is, graphs with bounded tree-width, contains trees, outerplanar graphs, series-parallel graphs, etc. A series-parallel graph is indeed a partial 2-tree. In this section we show that the results in Sections 3 and 4 can be extended to partial k-trees.

For a bounded positive integer k, a *k-tree* is recursively defined as follows [1,2]:

(1) A complete graph with k vertices is a k-tree.
(2) If $G = (V, E)$ is a k-tree and k vertices $v_1, v_2, \ldots, v_k$ induce a complete subgraph of G, then $G' = (V \cup \{w\}, E \cup \{(v_i, w) : 1 \le i \le k\})$ is a k-tree where w is a new vertex not contained in G.

Any subgraph of a k-tree is called a *partial k-tree*.

A binary tree $T = (V_T, E_T)$ is called a *tree decomposition* of a partial k-tree $G = (V, E)$ if T satisfies the following conditions (a)–(e):

(a) every node $X \in V_T$ is a subset of V and $|X| = k + 1$;
(b) $\bigcup_{X \in V_T} X = V$;
(c) for each edge $e = (u, v) \in E$, T has a leaf $X \in V_T$ such that $u, v \in X$;
(d) if node X_p lies on the path in T from node X_q to node X_r, then $X_q \cap X_r \subseteq X_p$; and
(e) each internal node X_i of T has exactly two children, say X_ℓ and X_r, such that $|X_\ell - X_r| = 1$ and either $X_i = X_\ell$ or $X_i = X_r$.

Each node X of T corresponds to a subgraph G_X of G. If X is a leaf of T, then G_X is a subgraph of G induced by the vertices in X. If X_ℓ and X_r are the two children of an internal node X_i of T, then G_{X_i} is a union of G_{X_ℓ} and G_{X_r}, whose common vertices are all contained in X_i. Thus $G = G_{X_{root}}$ for the root X_{root} of T.

One can easily observe from the definitions above that $\chi(G) \le k + 1$ for every partial k-tree G. Therefore, by Lemma 2.1 we have $\chi_b(G) \le (2k + 1)B$. Similarly as in Section 3, we compute the counterparts of χ_{ij} from leaves to the root of a tree decomposition T of G. Since $\chi_b(G) \le (2k + 1)B$ and $|X| = k + 1$ for every node X of T, there are a number $((2k + 1)B)^{k+1}$ of counterparts of χ_{ij}. Since T has $O(n)$ leaves, the counterparts of χ_{ij} can be computed in time

$$(k + 1)^2 \times ((2k + 1)B)^{k+1} \times n = O(B^{k+1}n)$$

for all leaves of T. Since T has $O(n)$ nodes and every internal node X_i of T has two children X_ℓ and X_r such that $|X_\ell - X_r| = 1$ and either $X_i = X_\ell$ or $X_i = X_r$, the counterparts of χ_{ij} can be computed in time

$$((2k + 1)B)^{k+1} \times (2k + 1)B \times n = O(B^{k+2}n)$$

for all internal nodes X_i of T. From the counterparts of χ_{ij} for the root of T, $\chi_b(G)$ can be computed in time $O(B^{k+1})$. Thus $\chi_b(G)$ can be computed in time $O(B^{k+2}n)$. Since the time is bounded by a polynomial in n and B, by Theorem 4.1 the scheme in Section 4 is an FPTAS and takes time

$$B_\sigma{}^{k+2} n = O\left(\left(\frac{n}{\varepsilon}\right)^{k+2} n\right).$$

We thus have the following corollary.

Corollary 5.1. *There is a fully polynomial-time approximation scheme for the b-coloring problem on partial k-trees.*

6. Conclusions

We first investigated the fundamental properties of a b-coloring. We then gave a pseudo polynomial-time exact algorithm and a fully polynomial-time approximation scheme for the b-coloring problem on series-parallel graphs and partial k-trees. It is desired to improve the time complexities. It is open whether the b-coloring problem can be solved in polynomial time or is NP-hard for series-parallel graphs or partial 2-trees.

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